Integration of SGD regulatory and expression data into the GRNmap and GRNsight applications for modeling and visualizing small-to-medium gene regulatory networks

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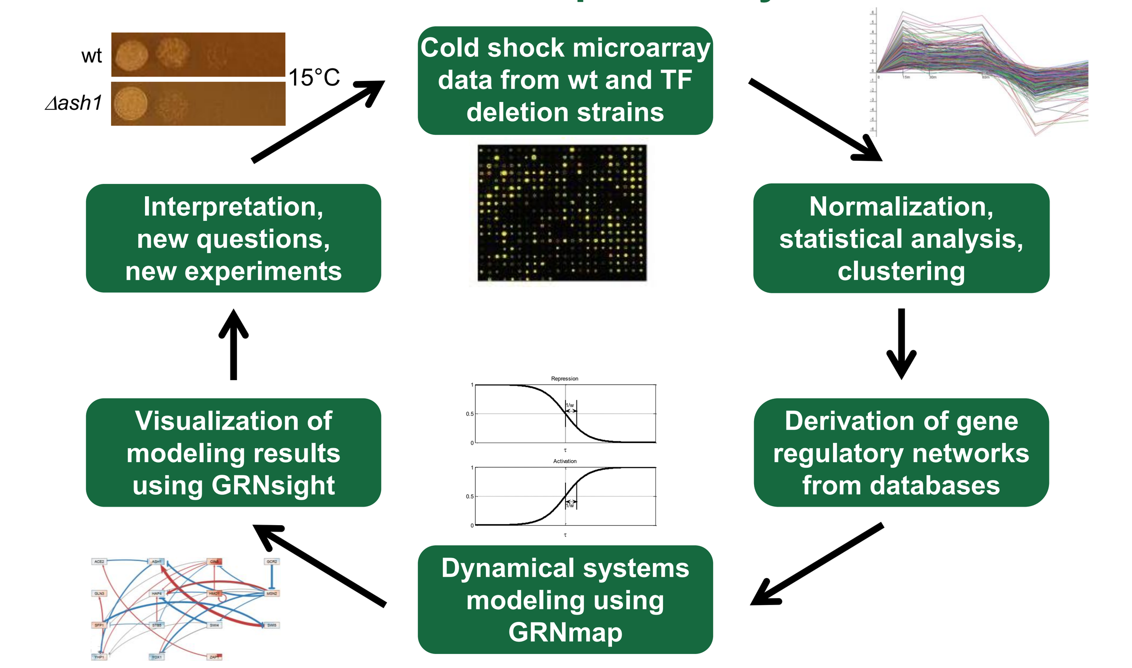
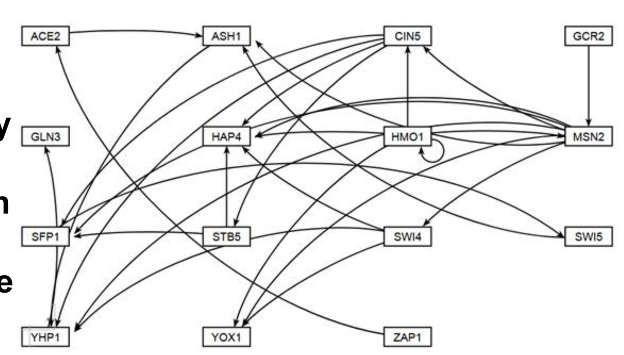
http://kdahlquist.github.io/GRNmap/ and http://dondi.github.io/GRNsight/

We have previously used a systems biology approach to understand the regulation of the cold shock response in yeast



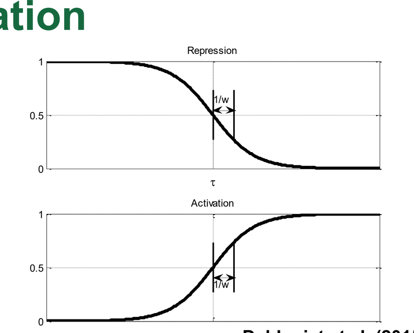
A "medium-scale" gene regulatory network that regulates the cold shock response

- Each node represents one gene encoding a transcription factor.
- Each edge represents a regulatory relationship, either activation or repression, depending on the sign of the weight.
- This network was derived from the YEASTRACT database.

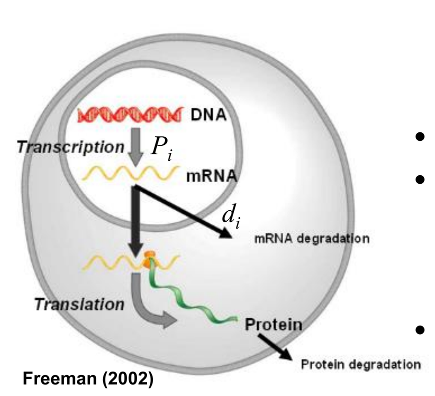


GRNmap: Gene Regulatory Network modeling and parameter estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1+\exp\left(-\left(\sum_j \left(w_{ij}x_j(t)\right)-b_i\right)\right)} - d_i x_i(t)$$



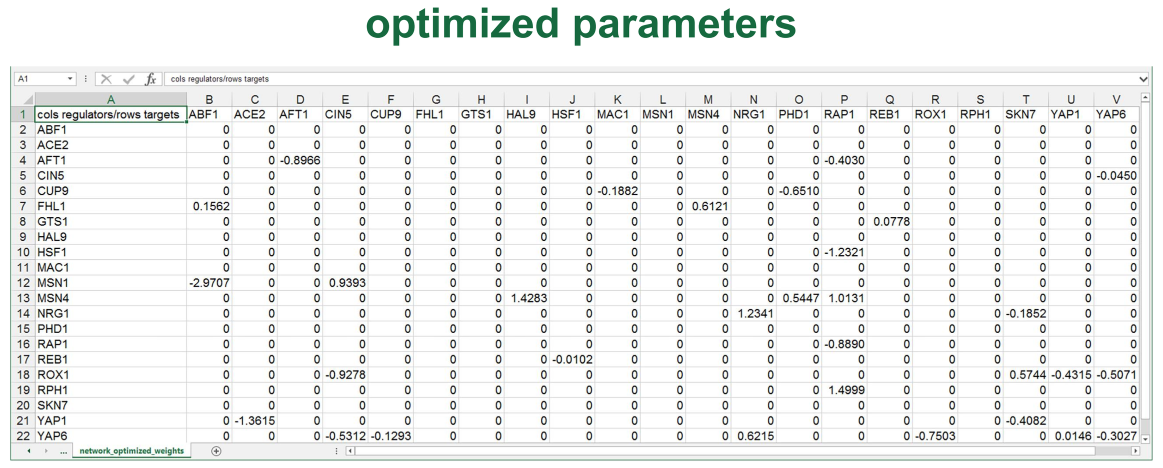
Dahlquist et al. (2015)



Freeman (2002)

- Model is coded in MATLAB.
- Input Excel workbook contains:
  - Network adjacency matrix
  - Timecourse expression data
  - mRNA degradation rates
- Output workbook contains optimized:
  - Production rates, P
  - Threshold b
  - Weight parameters, w, give the direction (activation or repression) and magnitude of the regulatory relationship

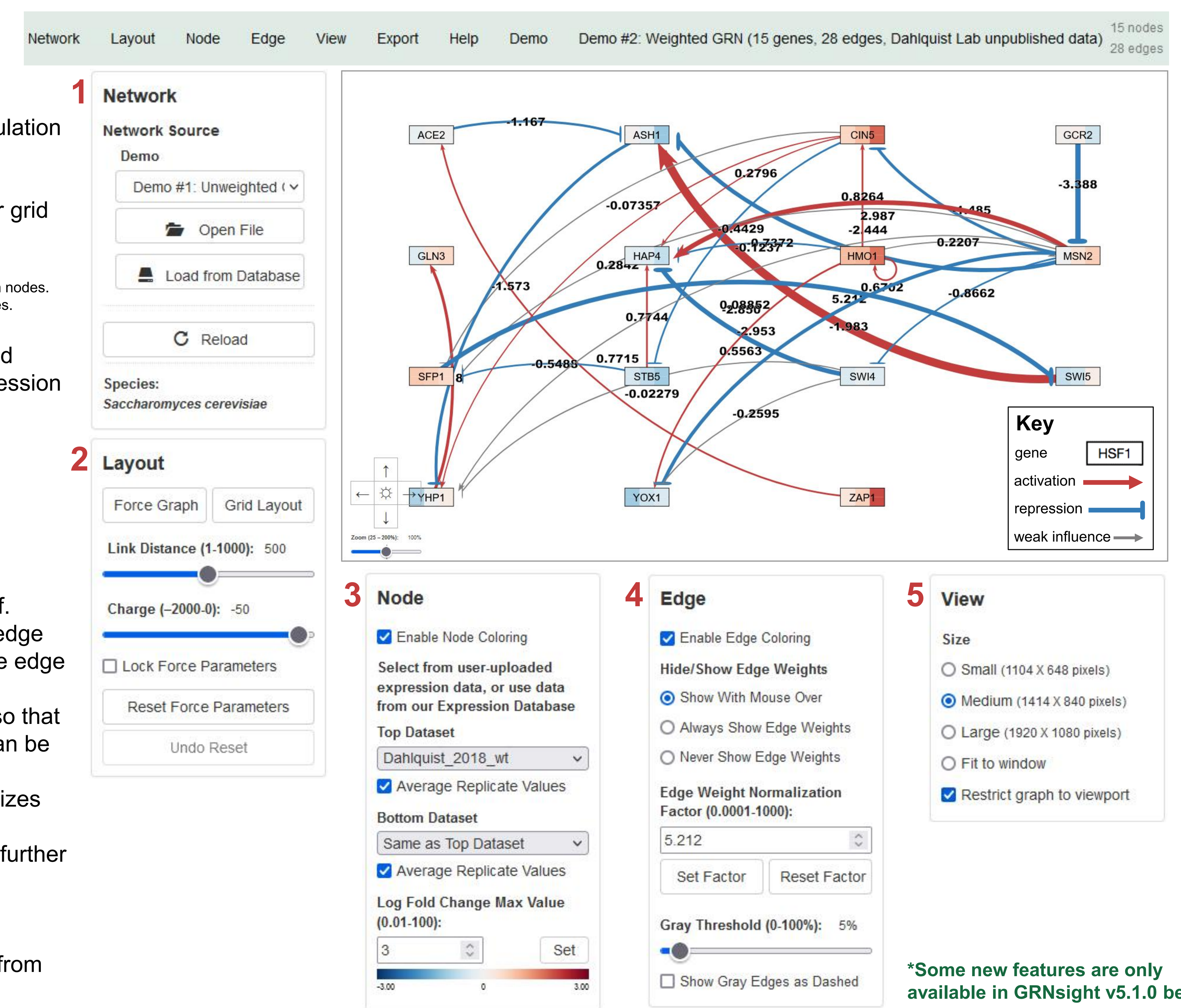
GRNmap produces a Excel workbook with optimized parameters



- Screenshot of the Excel worksheet containing the optimized regulatory weights in the form of an adjacency matrix.
- This format is directly read by GRNsight.

GRNsight, automatically lays out the network graph, coloring the edges based on weight values, and the nodes based on experimental or simulated expression data

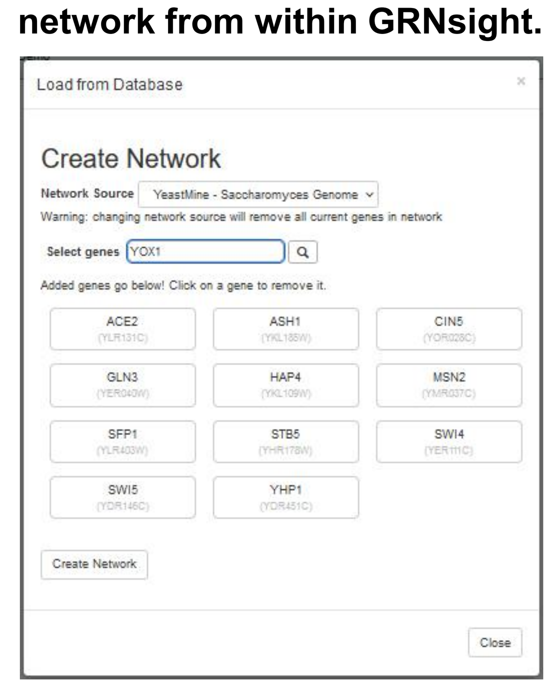
1. **Network**
   - Demo files are provided.
   - Can import and export Excel, SIF, or GraphML files.
   - Can construct a network from SGD regulation data stored in a backend database.
2. **Layout**
   - Users can select a force graph layout or grid layout.
   - Force graph parameter sliders:
     - Link distance determines the minimum distance between nodes.
     - Nodes have a charge which repels or attracts other nodes.
3. **Node**
   - Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
   - Datasets available:
     - Apweiler et al. 2012, GSE33098
     - Barreto et al. 2012, GSE24712
     - Dahlquist et al. 2018, GSE83656
     - Kitagawa et al. 2002, GSE9336
     - Thorsen et al. 2007, GSE6068
4. **Edge**
   - Edge coloring can be toggled on and off.
   - Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover.
   - Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
   - Changing the gray threshold deemphasizes weaker edges in the visualization.
   - Presenting gray edges as dashed lines further deemphasizes weak edges.
5. **View**
   - Multiple viewport sizes available.
   - Graph bounding box can be separated from viewport.
   - Zooming and scrolling available.



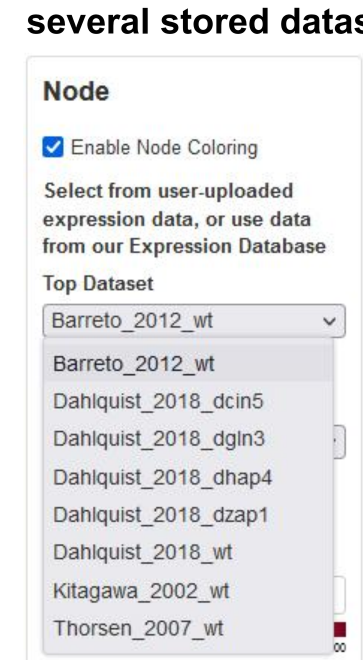
*Some new features are only available in GRNsight v5.1.0 beta.

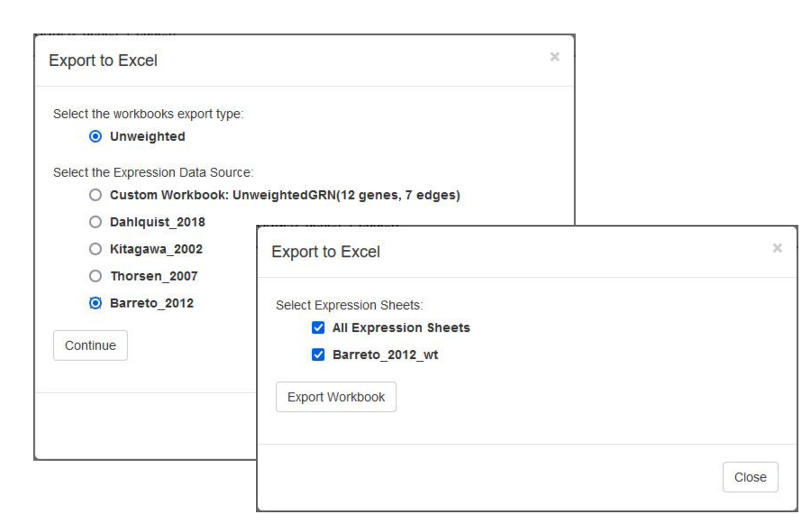
However, creating GRNmap or GRNsight input Excel workbooks by hand was time-consuming, error-prone, and was a rate-limiting step in analysis

- Thus, we created a backend database for GRNsight with SGD regulation data gathered from YeastMine.
- Users can now select genes that they want to include in the network from within GRNsight.



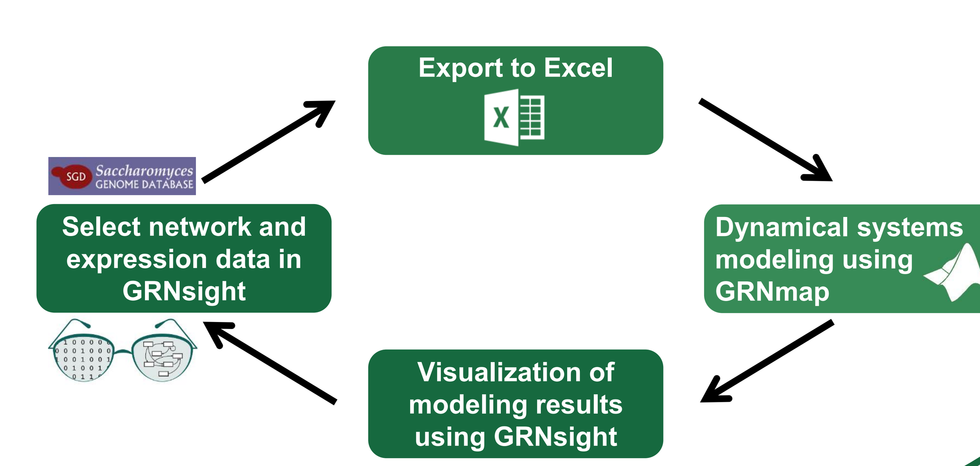
- We also created a backend expression database for GRNsight.
- With assistance from SGD staff, microarray datasets with timecourse data were identified from the SPELL compendium.
- Users can now color nodes with data from several stored datasets.



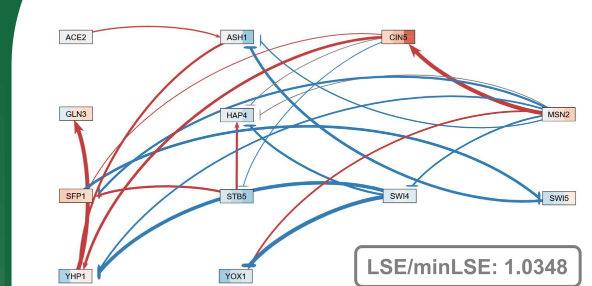


- Users can then export both the network and expression data to a GRNmap-compatible Excel workbook to perform modeling.

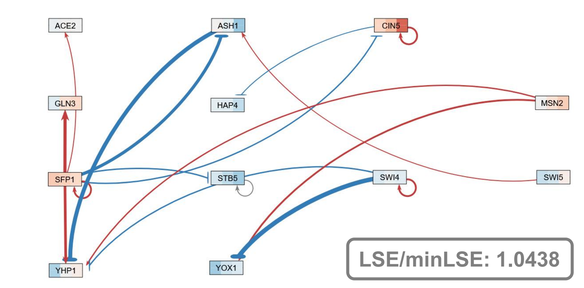
New Workflow



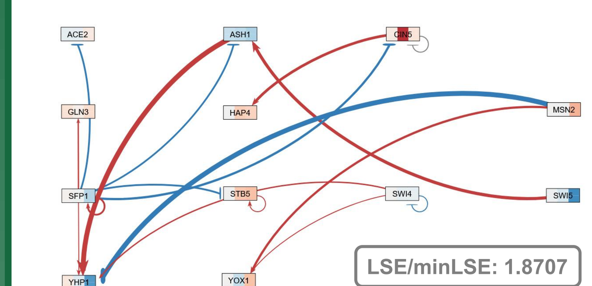
Improved workflow facilitates comparisons between models



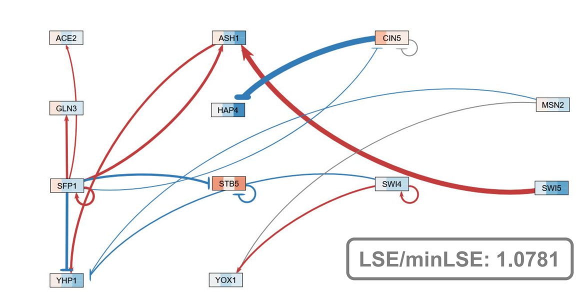
- 12-gene, 21 edge network from YEASTRACT
- Modeled with Dahlquist et al. (2018) expression data
- wt yeast subjected to 15, 30, and 60 minutes of cold shock at 13°C



- 12-gene, 17 edge network from SGD
- Modeled with Dahlquist et al. (2018) expression data
- wt yeast subjected to 15, 30, and 60 minutes of cold shock at 13°C



- 12-gene, 17 edge network from SGD
- Modeled with Kitagawa et al. (2002) expression data
- wt yeast treated with 75 μM thiuram for 15, 30, and 120 minutes



- 12-gene, 17 edge network from SGD
- Modeled with Thorsen et al. (2007) expression data
- wt yeast treated with 1 mM As(III) for 15, 30, and 60 minutes

Discussion

- Gene regulatory networks created from SGD data have fewer edges than those derived from YEASTRACT.
  - The GRNs shown in the left and center panels derived from YEASTRACT have 15 genes and 28 edges.
  - When building the network from SGD data, GCR2, HMO1, and ZAP1 were no longer connected, resulting in a 12-gene, 17-edge network.
- The new workflow eliminates manual manipulation of Excel workbooks, speeding up the rate of research.
- Edge weight normalization allows the edge thickness to be fairly compared between graphs. (Normalization factor = 5.498)
- Nodes are colored with stripes, representing expression of that gene at each timepoint in the data.
- Least Squares Error (LSE) to minimum theoretical LSE (min LSE) is an overall measure of how well the parameters fit the data.
  - In this set of models, the YEASTRACT network with Dahlquist et al. (2018) data had the smallest LSE/minLSE ratio.

Future Directions

- We plan to expand the number of datasets available in the backend expression database.
- We plan to further model a large number of small gene regulatory networks to learn the network identities and features that explain patterns of gene expression in yeast.

Availability

- GRNsight is free and open to all users at http://dondi.github.io/GRNsight/, and there is no login requirement.
- GRNsight code is available under the open source BSD license at our GitHub repository at https://github.com/dondi/GRNsight.
- GRNmap MATLAB code and executable can be downloaded from http://kdahlquist.github.io/GRNmap/downloads.html under the BSD license.
- Some new GRNsight features are only available in GRNsight v5.1.0 beta.

Acknowledgements

We would like to thank the previous developers on the GRNmap and GRNsight projects as well as the students enrolled in the Fall 2019 Biology 367 Biological Databases course who piloted creating the expression database with the Barreto et al. (2012), Kitagawa et al. (2002), and Thorsen et al. (2007) datasets. We would especially like to thank Edith Wong from SGD for providing timecourse expression data from SPELL.

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